

1 Summary

Model (Run-name): model1 (test)

Iteration: 24

TAC: 154871.33474635327

Took: 4.535146474838257s

Total Time: 41.73745656013489s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

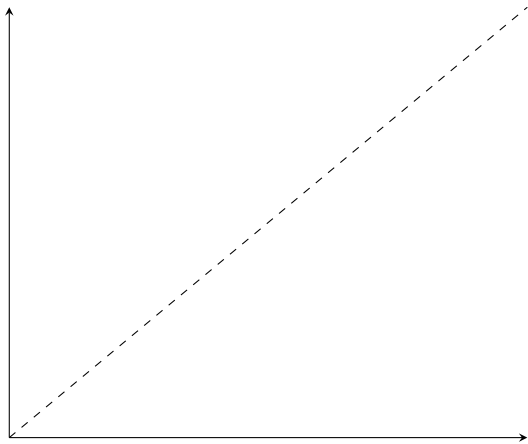
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

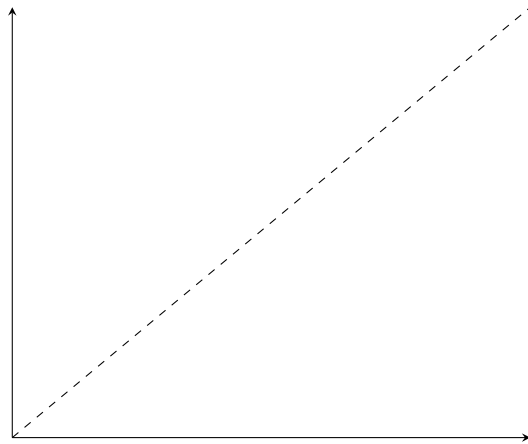
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05234600709175324	$b_{1,1,1}^{H,out}$	893.5589570013279
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017559819245540724	$b_{1,2,2}^{H,out}$	3736.8880006242584
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.02943852276867638	$b_{2,1,2}^{H,out}$	7923.59603228337
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014559904600459125	$b_{1,1,1}^{C,in}$	5986.696661644737
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009591089624654738	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02257499274638778	$b_{2,1,2}^{C,in}$	4164.084379739776
$A_{1,1,1}^\beta$	70.65581083979713	$q_{1,1,1}$	676.2067782712752	$b_{1,1,1}^{C,out}$	6662.903439916012
$A_{1,2,2}^\beta$	68.48329505760876	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	143.30274961189664	$q_{2,1,2}$	2435.689832593087	$b_{2,1,2}^{C,out}$	6599.774212332863
$A_{cu,1}^\beta$	5.0359974771272995	$q_{cu,1}$	173.7932217287248		
$A_{cu,2}^\beta$	37.65435035996605	$q_{cu,2}$	1964.310167406913		
$A_{hu,1}^\beta$	13.211215176722128	$q_{hu,1}$	488.1033891356376		
$A_{1,1,1}$	70.65581083979713	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	68.48329505760876	$t_{1,2}^{(H)}$	582.3793221728724		
$A_{2,1,2}$	143.30274961189664	$t_{1,3}^{(H)}$	387.3793221728725		
$A_{cu,1}$	5.0359974771272995	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.65435035996605	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.211215176722128	$t_{2,3}^{(H)}$	468.2155083703457		
$dt_{1,1,1}$	32.54022594237584	$t_{1,1}^{(C)}$	617.4597740576241		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.3793221728724		
$dt_{1,2,2}$	82.37932217287248	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.37932217287248	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.62067782712751	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.215508370345646	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.37932217287248	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.21550837034565	$t_{1,2,2}^H$	382.4080179515642		
$dt_{hu,1}$	62.54022594237584	$t_{2,1,2}^H$	451.27837189025587		
$f_{1,1,1}^H$	2.415024208111697	$t_{1,1,1}^C$	631.4149834951634		
$f_{1,2,2}^H$	9.764978519300461	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	17.558111635383824	$t_{2,1,2}^C$	649.878468329369		
$f_{1,1,1}^C$	10.460449669082664	$b_{1,1,1}^{H,in}$	1569.765735272603		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5686.888000624258		
$f_{2,1,2}^C$	10.156303365218964	$b_{2,1,2}^{H,in}$	10359.285864876456		

3 *RecLMTD* Tangent Points

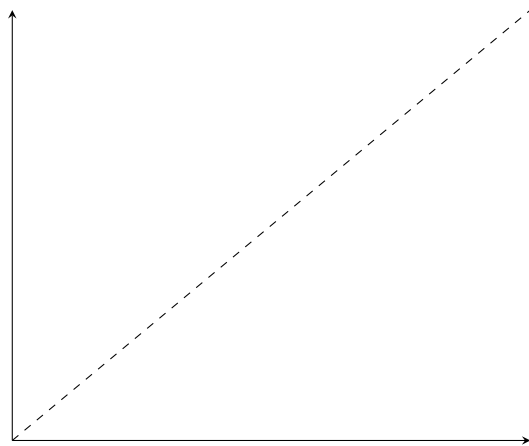
- $(1, 1, 1)$
 - Value: 0.052346
 - Error: 0.000000049922775
 - Relative error: 0.000000953706511
- $(1, 2, 2)$
 - Value: 0.017560
 - Error: 0.000000553920539
 - Relative error: 0.000031543779521
- $(2, 1, 2)$
 - Value: 0.029439
 - Error: 0.000000661510106
 - Relative error: 0.000022470395227
- $cu, 1$
 - Value: 0.014560
 - Error: 0.000000093387076
 - Relative error: 0.000006413948388
- $cu, 2$
 - Value: 0.009591
 - Error: 0.000000002037882
 - Relative error: 0.000000212476499
- $hu, 1$
 - Value: 0.022575
 - Error: 0.000000529145432
 - Relative error: 0.000023438901400



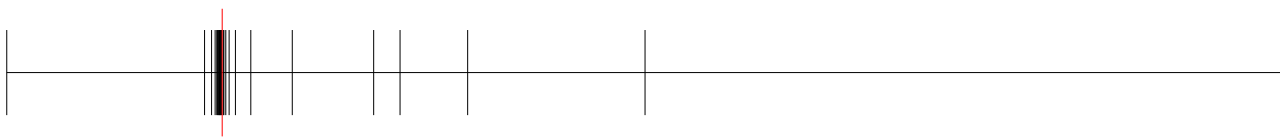
(a) *RecLMTD* - (1, 1, 1)



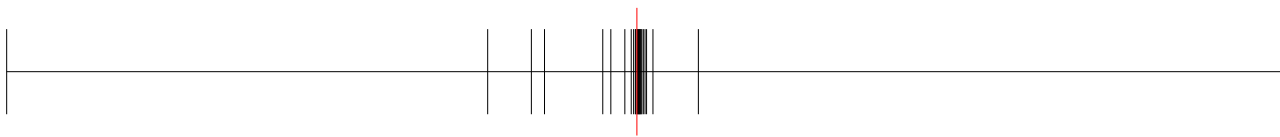
(b) *RecLMTD* - (1, 2, 2)



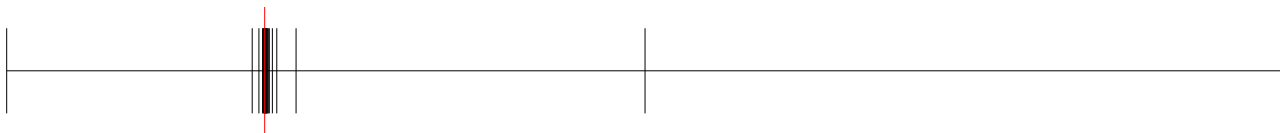
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1569.765735272603
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5686.888000624258
 - Absolute error: 0.03357047860390594
 - Relative error: -5.9031020885690946e-06
- (2, 1, 2)
 - Value: 10359.285864876456
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 893.5589570013279
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3736.8880006242584
 - Absolute error: 2.6819197189688566
 - Relative error: 0.0007182034576727684
- (2, 1, 2)
 - Value: 7923.59603228337
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5986.696661644737
 - Absolute error: 0.6484295682457741
 - Relative error: -0.00010830001584465346
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4164.084379739776
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6662.903439916012
 - Absolute error: 58.01878476019465
 - Relative error: 0.008784223766103893
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6599.774212332863
 - Absolute error: 0.5886625440534772
 - Relative error: -8.918639099285198e-05



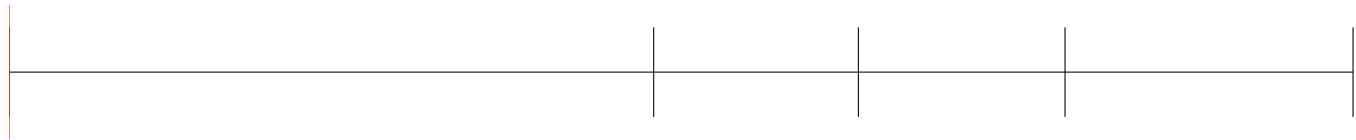
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



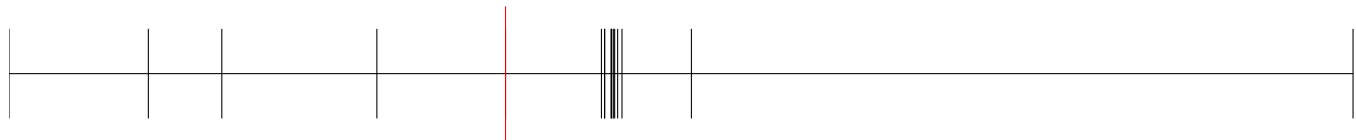
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



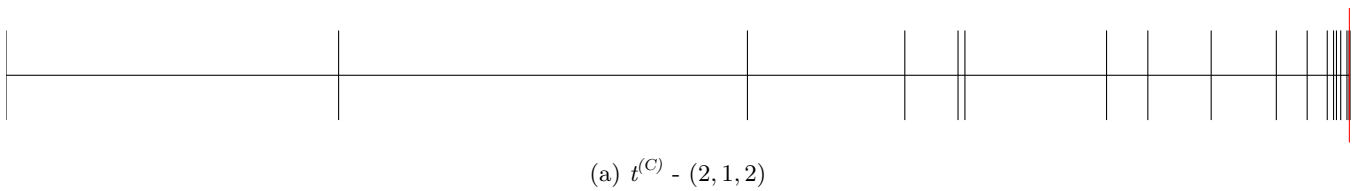
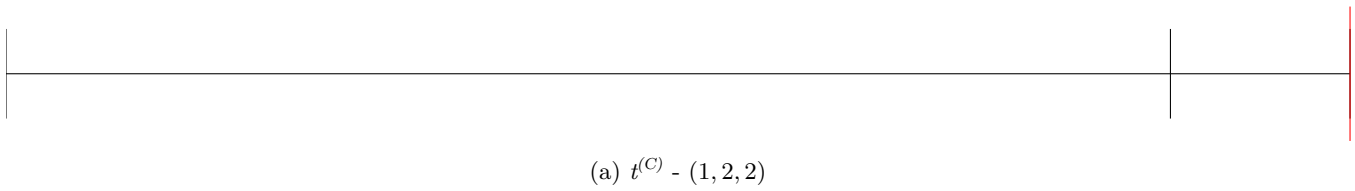
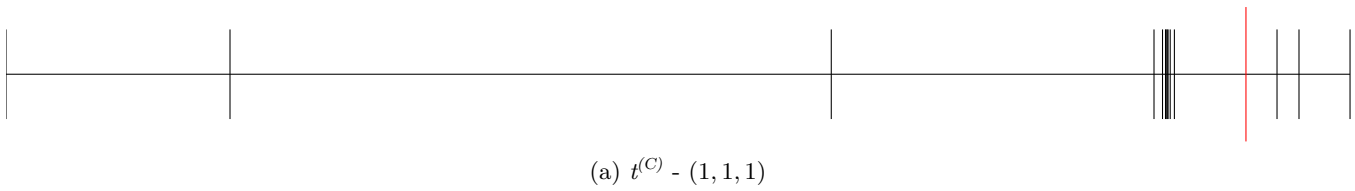
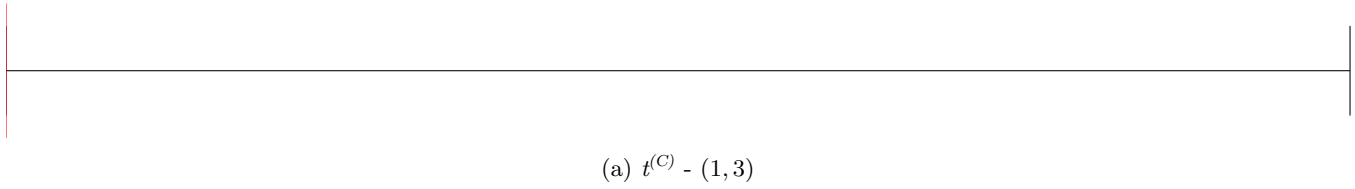
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

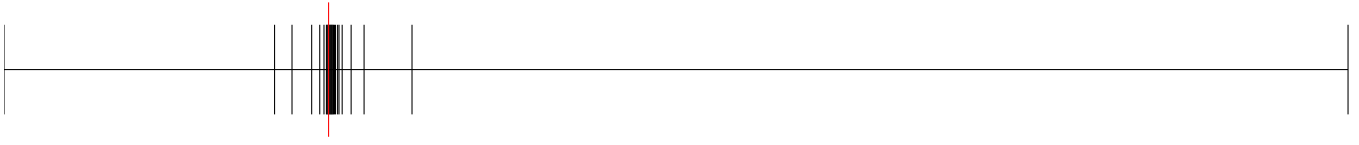
- $(1, 1, 1)$
 - Value: 70.655811
 - Error: 0.137638781962437
 - Relative error: 0.001944230471856
- $(1, 2, 2)$
 - Value: 68.483295
 - Error: 0.0000000000000071
 - Relative error: 0.0000000000000001
- $(2, 1, 2)$
 - Value: 143.302750
 - Error: 0.103471576553659
 - Relative error: 0.000721527808879

5.2 A_{CU}

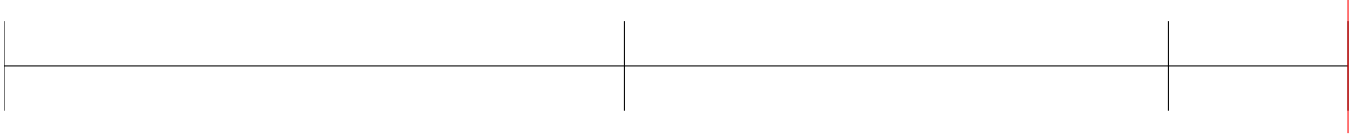
- $cu, 1$
 - Value: 5.035997
 - Error: 0.024827980026046
 - Relative error: 0.004905915099473
- $cu, 2$
 - Value: 37.654350
 - Error: 0.025399372474460
 - Relative error: 0.000674085487691

5.3 A_{HU}

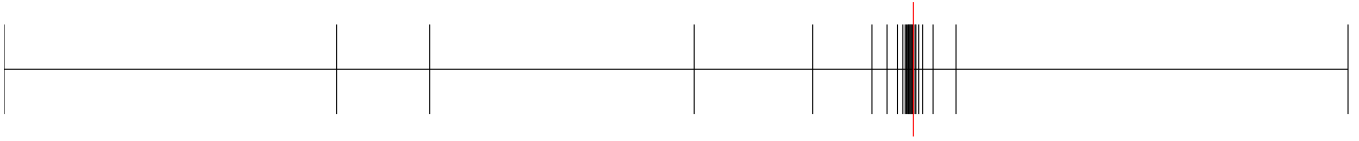
- $hu, 1$
 - Value: 13.211215
 - Error: 0.011501386347044
 - Relative error: 0.000869820228860



(a) $A^\beta - (1, 1, 1)$



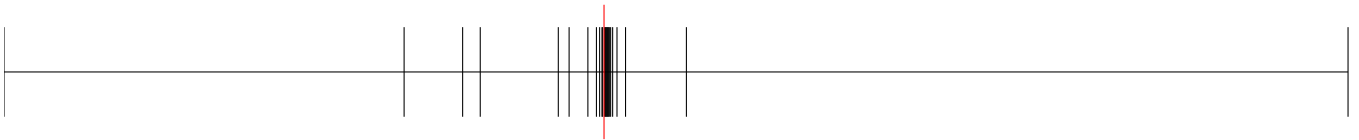
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

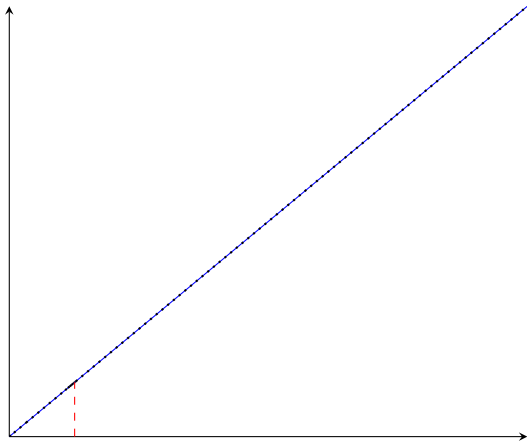


(a) $A^\beta - hu, 1$

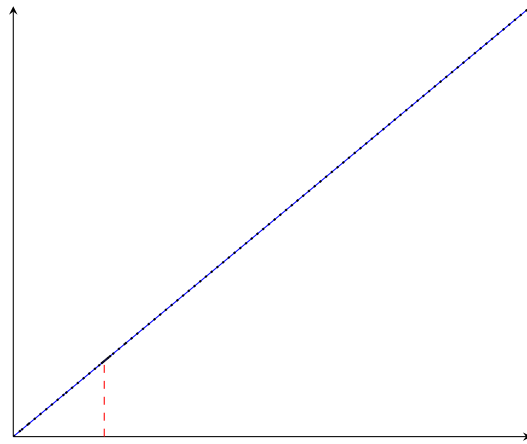
6 A^β Breakpoints

- (1,1,1)
 - q : 70.655811
 - $\hat{q}^\beta$: 70.655811
 - q^β : 70.655811
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, **70.60541450524272**, **70.83181816751056**, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,2,2)
 - q : 68.483295
 - $\hat{q}^\beta$: 68.483295
 - q^β : 68.483295
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, **68.10969007738701**, **68.66676327863097**, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 143.302750
 - $\hat{q}^\beta$: 143.302750
 - q^β : 143.302750
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, **142.4338700260237**, **144.67005099128556**, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 5.035997
 - $\hat{q}^\beta$: 5.035997
 - q^β : 5.035997
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335, 4.974201802143407, 4.983938626310074, **4.993068227518925**, **5.0418124100090695**, 5.081618324442152, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

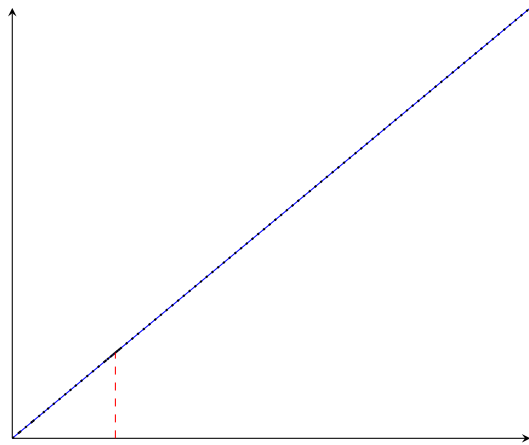
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.654350
 - $\hat{q}^\beta$: 37.654350
 - q^β : 37.654350
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, **37.612770977560544**, **37.6551299822246**, 37.682669717644174, 37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.211215
 - $\hat{q}^\beta$: 13.211215
 - q^β : 13.211215
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.189852638760229, **13.192252946941991**, **13.213749911872785**, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



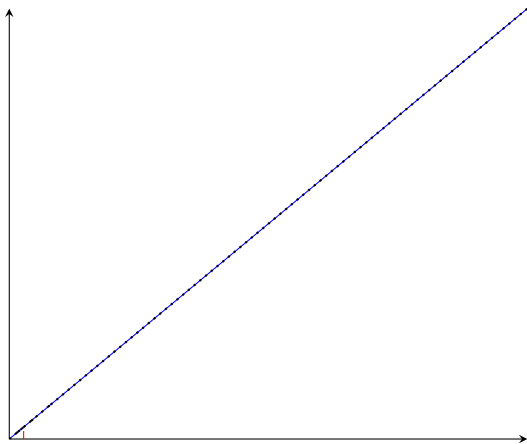
(a) $q^\beta - (1, 1, 1)$



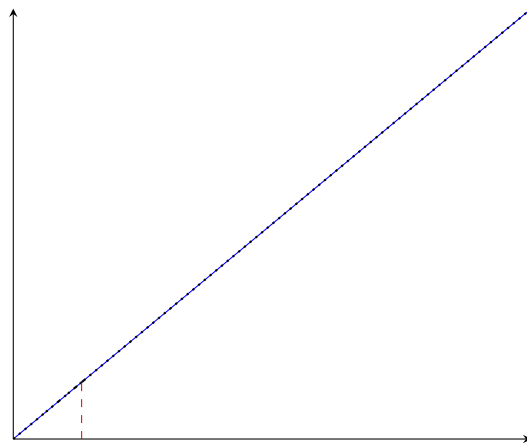
(b) $q^\beta - (1, 2, 2)$



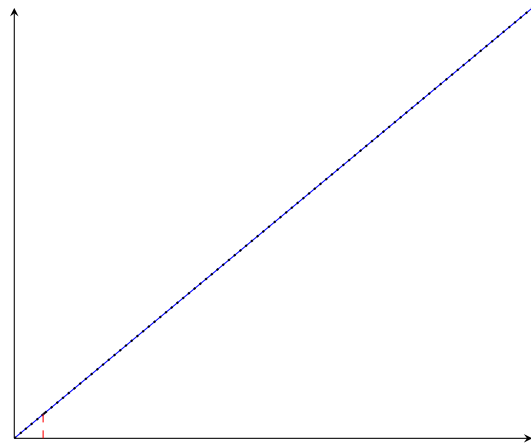
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.06	$b_{2,2,2}^{H,out}$	1440.7141351235439
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.06	$b_{1,1,2}^{C,in}$	1985.9156202602248
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.16	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.016070473816068356	$b_{2,1,1}^{C,in}$	2598.99317094835
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.16	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038186	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.0043402777777777596	$b_{1,1,2}^{C,out}$	1985.9156202602248
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.0055493298268132556	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.06	$b_{2,1,1}^{C,out}$	2598.99317094835
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.06	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.06	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	563.3974443535717		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	578.5046759713259		
$dt_{1,2,1}$	280.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	556.2752414439839		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.011203584752		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	136.90522110446636		
$f_{1,1,2}^H$	0.23502148069953796	$b_{1,2,1}^{H,in}$	4930.234264727397		
$f_{1,2,1}^H$	7.5849757918883025	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	1440.7141351235439		
$f_{2,2,2}^H$	2.441888364616176	$b_{1,1,2}^{H,out}$	136.90522110446636		
$f_{1,1,2}^C$	4.843696634781036	$b_{1,2,1}^{H,out}$	4930.234264727396		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.539550330917336	$b_{2,2,1}^{H,out}$	0.0		